

Evaluation of Case Study: Predicting Parts Utilization in Hardware Support

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Introduction

What problem was being solved?

The core problem for this major U.S. hardware manufacturer was the inefficiency of their field service operations. When customers called the help desk, the company needed to predict whether a replacement part would be required to fix the device. Without this insight, engineers often arrived at customer sites without the necessary hardware, leading to costly second trips. The project specifically focused on the “PartsUsed” target variable to determine if a part was needed and which part was the most likely candidate for the repair.

Why this problem was important to solve?

Field service is a massive cost and any reduction in second trips represents a strategic corporate asset. For this firm, the goal was not to classify every single ticket perfectly but to identify at least 20 percent of the tickets that needed a part with high accuracy. Solving this meant they could ensure the right parts were on the service trucks, which improved the first-time fix rate, saved significant labor costs, and protected the company from revealing too much operational detail to its competitors.

How was the data acquired?

The data was massive and high quality, consisting of over 1,000,000 support tickets. Each ticket included variables like the call timestamp, business name, country, and device error codes. The most valuable data point was the unstructured problem description transcribed during the help desk calls. The team linked these descriptions to the actual outcome of the ticket to create a labeled dataset for the modeling process.

Methods and Results

What steps were taken to prepare the data?

Preparation involved extensive cleaning of codes and human-transcribed text. A database programmer manually fixed misspellings and abbreviations, essentially stemming words like STCK and STICKING into a single keyword called STICK. They identified over 600 keywords and phrases. The real breakthrough came when they realized that binary 1/0 dummy variables were not enough. They created new features based on a temporal framework, using the historical percentage of times a keyword resulted in a part being used over a specific prior time period. This allowed the models to account for trends and the actual experience of engineers over time.

How was this problem solved?

The problem was solved using an unconventional approach to decision trees. When standard trees failed to meet the overall accuracy requirements, the team stopped looking at the trees as a single solution. Instead, they treated them as feature creators to find interesting business rules. They built more than 10,000 trees and collected over 20,000 terminal nodes that showed a high likelihood of a part being needed. By treating each terminal node as an independent rule, they created a sequence that could be applied to incoming tickets.

What modeling techniques were used?

Decision trees were the primary tool because they handle wide, categorical data efficiently and are easy for humans to interpret. The team also explored Random Forests but found them unsuitable because the individual branches were overfit and the logic was too hard to explain to the support engineers. They also considered association rules because they are exhaustive, though these would have required more extensive binning of the percentage data.

Why did the team choose the methods/models they did?

The team chose decision trees for their transparency and scalability. Support engineers needed to understand why a part was being recommended before they drove to a customer site. While ensemble methods like Random Forests were more accurate, they lacked the clarity required for deployment. The decision to use a rule-based sequence allowed the company to keep the logic simple and easy to communicate while still hitting the accuracy targets for the most critical tickets.

What metrics were used to evaluate the results?

The primary evaluation metric was the PartsUsed percentage within specific terminal nodes. They also tracked the lift over the baseline rate of 23.2 percent. A unique metric used in the final stage was the PartsUsed Percentage in Sequence. This was chosen because it showed how a rule performed after previous rules in the sorted list had already fired. This helped the team prune out redundant rules that only matched a handful of tickets or had dropping accuracy in the real-world sequence.

Conclusion

How were the results or model implemented?

The project was implemented as a sorted list of rules that fired in a specific sequence from the highest likelihood to the lowest. Because these rules were essentially "if-then" conditions, they were easily converted into SQL. This made the technical deployment into the existing corporate infrastructure very simple. When a new ticket came in, it ran through the sequence until it matched a rule, giving the engineer a clear prediction on whether to bring a part.

What were the actionable consequences of the case study?

The results were so successful that the modeling was expanded to cover more tickets and even higher accuracy scores. The cost savings were significant due to the reduction in unnecessary service

trips. The company also gained a deeper understanding of their customer base, realizing that high-expertise clients actually had higher part-usage rates because they solved the easy non-hardware problems themselves.

What did the team learn from the case study?

The biggest takeaway was that domain expertise and feature engineering are more important than just having a lot of data. Without the new temporal features created by the subject matter experts, the performance would have stayed below the business requirements regardless of the algorithm used. They also learned that a model does not have to be a perfect black box to be successful; sometimes picking the best rules from thousands of trees is the more practical path to deployment.

How should or would the team approach the problem differently in the future?

In the future, the team would likely lean even harder into the idea of trees as rule generators rather than trying to find one single perfect tree. They saw that the iterative nature of the project—going back to the business understanding and asking how an informed person thinks about the data—was what finally led to the win. They would also likely automate more of the temporal feature creation to stay ahead of changing hardware trends and keep the rules fresh.

References

Abbott, D. (2014). Applied predictive analytics: Principles and techniques for the professional data analyst. John Wiley & Sons.